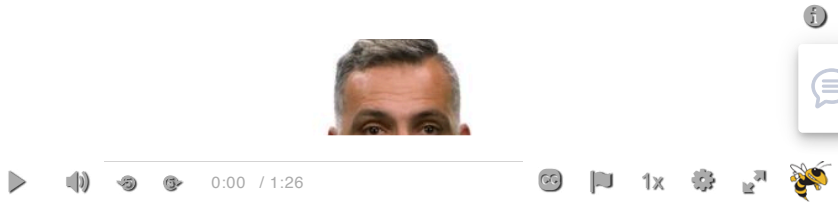


Lesson Twelve Overview



Throughout the lecture, we are giving you some “food for thought” -- these are questions we suggest you think about as you go through this lesson. You do not need to submit solutions.

Learning Objectives

Students will be able to:

- Appreciate the value of modeling in the context of network science
- Become familiar with some commonly used models such as preferential attachment or stochastic block matrix models
- Understand the differences between static vs dynamic models, and stochastic vs optimization-driven models
- Understand the value of network modeling through case studies

Required and Recommended Reading

Required Reading

- Chapter-5 from A-L. Barabási, [Network Science](http://www.networksciencebook.com/) (<http://www.networksciencebook.com/>), 2015.
- Chapter-18 (mostly sections 18.3 and 18.7) from D. Easley and J. Kleinberg, [Networks, Crowds and Markets](http://www.amazon.com/Networks-Crowds-Markets-Reasoning-Connected/dp/0521195330) (<http://www.amazon.com/Networks-Crowds-Markets-Reasoning-Connected/dp/0521195330>), Cambridge Univ Press, 2010 ([Online](https://www.cs.cornell.edu/home/kleinber/networks-book) (<https://www.cs.cornell.edu/home/kleinber/networks-book>)).
- [Hierarchical structure and the prediction of missing links in networks](https://www.nature.com/articles/nature06830)  (<https://www.nature.com/articles/nature06830>), by A.Clauset et al. Nature, 2008

Recommended Reading

- ["Heuristically Optimized Trade-Offs: A New Paradigm for Power Laws in the Internet"](https://link.springer.com/chapter/10.1007/3-540-45465-9_11),  (https://link.springer.com/chapter/10.1007/3-540-45465-9_11) by Fabrikant et al., 2002

Assignments

L12: Knowledge Check

L12: Quiz